

Aug. 4, 1931.

F. REGERO
WATCH CLEANING MACHINE

1,817,266

Filed Aug. 5, 1929

2 Sheets-Sheet 1

Fig. 1.

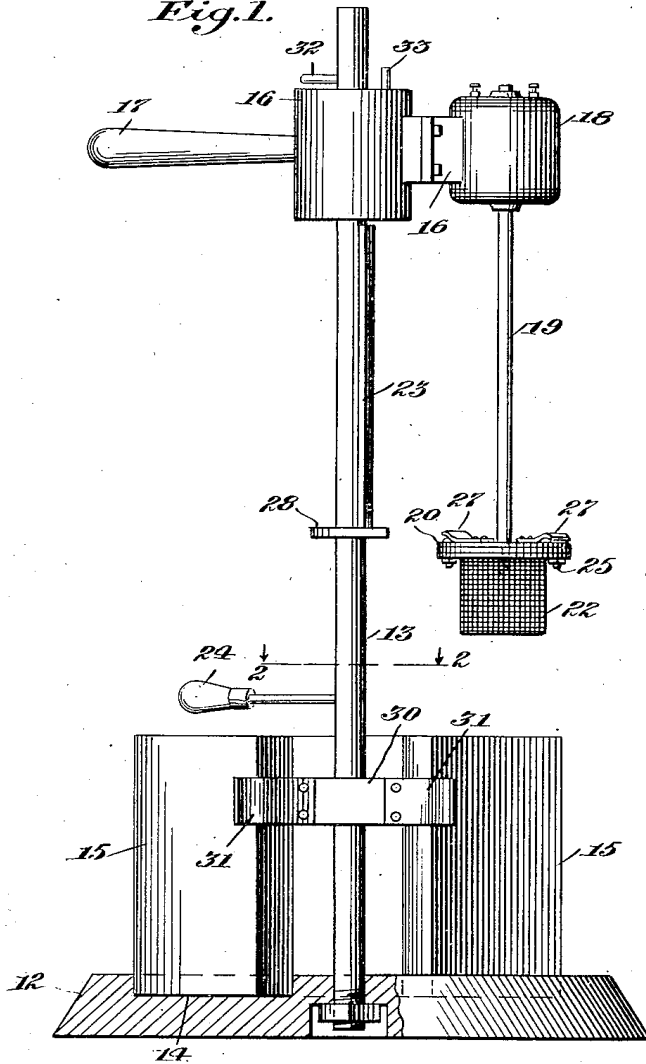


Fig. 4.

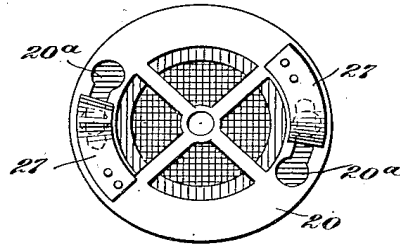
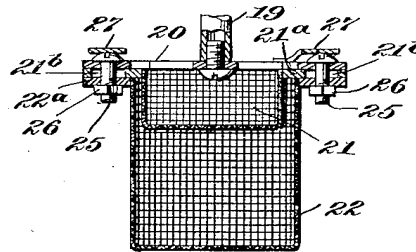


Fig. 5.



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2 Sheets-Sheet 2

Fig. 2.

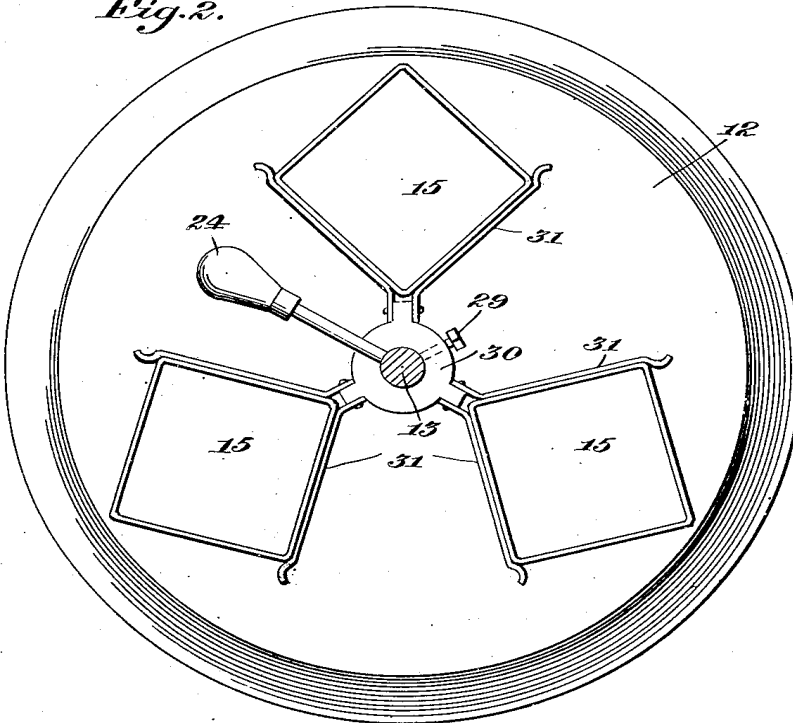
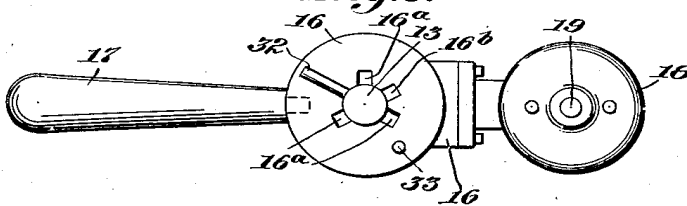


Fig. 3.



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UNITED STATES PATENT OFFICE

FRANK REGERO, OF MACON, GEORGIA, ASSIGNOR, BY DIRECT AND MESNE ASSIGNMENTS, TO L. & R. MANUFACTURING COMPANY, A CORPORATION OF NEW JERSEY

WATCH CLEANING MACHINE

Application filed August 5, 1929. Serial No. 383,748.

This invention has for its object to provide a watch cleaning machine which is comparatively simple in construction and convenient and efficient in operation.

5 In the accompanying drawings Fig. 1 is an elevation of the improved machine. Fig. 2 is a horizontal sectional view of the machine on line 2—2, Fig. 1. Fig. 3 is a plan view of the upper part of the machine. 10 Figs. 4 and 5 are detail views of the baskets to hold the parts to be cleaned.

Referring to the drawings, 12 denotes a base to which is fixed a standard 13. The base 12 is provided with a series of recesses 15 or pockets 14 in which tanks or jars 15, to hold cleansing and drying solutions, are located. Vertically movable on the standard 13 is a carrier 16 having a handle 17 by which it may be conveniently lowered or raised, 20 the said carrier supporting an electric motor 18 having a shaft 19 to the lower end of which is attached a head or disk 20 which supports upper and lower wire baskets 21 and 22. The smaller basket 21 is intended 25 for holding the smaller watch parts, and the larger basket 22 for holding the larger watch parts, but a single basket for holding all of the watch parts may be used, if desired.

The standard 13 is provided with a spline 30 or key 23 and the hub of the carrier 16 is provided with three vertical slots 16^a. In the operation of the machine the carrier may be partially rotated by the handle 17 to bring one of the slots 16^a into register with the spline or key 23, so as to permit the said carrier and the baskets 21 and 22 to be brought into register vertically, with any one of the tanks or jars 15 in which the said baskets will be rotated for a cleansing operation. The hub of the carrier 16 is also provided with a slot 16^b which may be caused to register with the spline or key 23 when it is desired to bring the baskets 21 and 22 over an electric light or heater 24 when it is desired to dry the parts which have been cleansed. This light or heater may be electrically connected with an electric light circuit from which the motor 18 may also be driven. 45

50 The basket 22 is provided with a flange

22^a to which are fixed bolts 25 held in place by set nuts 26, and the basket 21 is provided with a flange 21^a having large openings 21^b which may be passed over the heads of the bolts 25 in locating the said basket 21 in place. 55 The head or disk 20 is provided with button-hole or slotted openings 20^a the larger parts of which will permit the passage of the heads of the bolts 25, and then by giving a partial rotation either to the said head or disk, or 60 to the baskets, the said bolts will be caused to occupy the smaller portions of said slots or openings so as to properly attach the baskets to the said head or disk. With the parts thus disposed the heads of the bolts 25 will be brought beneath the springs 27, 65 attached to the head or disk 20, which springs will serve to frictionally retain or lock the baskets to the said head or disk. It will thus be understood that the baskets or parts-holding receptacles, one of which is 70 nested within the other, are removably mounted on the rotating head or disk 20.

When the carrier 16 and the parts supported thereby are in an elevated position, as shown in Fig. 1, the said carrier will be turned to such position that the slots 16^a and 16^b will be out of register with the key or spline 23 and the said key or spline will thus hold the carrier in its raised position. For clearness 75 of illustration the said carrier is shown as being raised slightly above said key or spline in Fig. 1, but it will be understood that the hub of said carrier will rest on said key or spline when in lifted position. When, however, the carrier is lowered to cause the baskets 21 and 22 to be dipped into the tanks or jars 15, which will be filled, or partly filled, with cleansing and drying solutions, the said carrier will rest against a stop 28 fixed to the 80 standard 13. In lifting the carrier 16 its upward movement will be limited by a suitable stop, as 32, on said standard 13. Also the carrier 16 may be provided with a stud 33 whereby the engagement with the stop 32 will 85 limit the turning movement of the said carrier.

Attached to the standard 13, by a set screw 29, is a hub 30 having three radial lugs to which are attached springs 31 which serve 90

to hold or steady the tanks or jars 15 in place, but permitting of their ready removal when desired.

In the operation of the machine, after a watch has been taken apart, the larger parts of the watch will be placed in the lower basket 22 and the smaller parts in the upper basket 21. The carrier 16 will then be partially rotated by its handle 17 to bring one of the slots 16^a in said carrier into register with the spline or key 23, and said carrier will then be lowered to cause the basket to be dipped into any one of the tanks or jars 15. The electric motor will then be started to cause the basket to be rotated in the solution in the jar into which the said baskets may be dipped, to cleanse the watch parts in the said basket. It is intended that one of the tanks or jars 15 will contain a strong cleansing solution and the other tanks or jars will contain different or drying solutions, and after the watch parts have been thoroughly cleansed the carrier will be lifted and partially turned to bring the slot 16^b in said carrier into register with the spline or key 23 so as to locate the baskets containing the watch parts over the electric light or heater 24 for a spinning drying operation.

From the foregoing it will be understood that the invention provides a convenient watch cleansing machine capable of easy and efficient operation for cleaning parts of a watch, or any other small parts, so that said parts may be thoroughly cleansed and dried in a very few minutes. The invention is not, however, to be understood as being limited to the specific constructions shown and described, as variations, within the province of mechanical skill, may be made without departing from the scope of the invention as defined by the claims hereunto appended.

Having thus described my invention, I claim and desire to secure by Letters Patent:—

1. In a watch-cleaning machine, the combination with a base, of a standard supported by said base, tanks, for holding solutions mounted on said base, a carrier vertically movable and adapted to be turned on said standard, an electric motor supported by said carrier and having a depending shaft, and nested baskets, for holding parts to be cleansed, connected to the lower end of said shaft.

2. In a watch-cleaning machine, the combination with a base, of a standard supported by said base, tanks, for holding solutions mounted on said base, a carrier vertically movable and adapted to be turned on said standard, an electric motor supported by said carrier and having a depending shaft, and means for holding parts to be cleansed, connected to the lower end of said shaft, said standard being provided with a spline and said carrier having vertical slots any one of which may be caused to register with said

spline when the parts to be cleansed are to be lowered into any one of said tanks.

3. In a watch-cleaning machine, the combination with a base, of a standard supported by said base, tanks, mounted on said base, a carrier vertically movable and adapted to be turned on said standard, an electric motor supported by said carrier and having a depending shaft, and means, for holding parts to be cleansed, connected to the lower end of said shaft, said standard being provided with spring-holding devices to retain said tanks in place but permitting of their ready removal.

In testimony whereof I affix my signature.
FRANK REGERO.

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